

Common Ownership’s Effect on Domestic Airline Productivity

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June 3, 2026

Abstract

This study investigates whether the rise of common ownership, where institutional investors hold stakes in multiple rival firms, influences firm-level productivity in the U.S. airline industry. Using quarterly data from DOT Form 41 and SEC Form 13F, we estimate Total Factor Productivity (TFP) using the Levinsohn and Petrin (2003) method and measure common ownership using the Modified Herfindahl-Hirschman Index. Economic theory presents contrasting predictions: common ownership could either reduce productivity by dampening competitive incentives or increase it through governance benefits. Our regression analysis finds no statistically significant relationship between common ownership and estimated TFP across all specifications. This result parallels findings in the pricing literature, suggesting that, for domestic airlines, common ownership may be uncorrelated with operational efficiency and cost reduction incentives after accounting for fixed effects and controls. The study extends the common ownership debate beyond pricing to the critical dimension of technological efficiency.

1 Introduction

1.1 Motivation and Background

Across U.S. firms, the share of equities held by institutional investors has risen from 33% to 61% over the past two decades. A direct consequence of this growth is the rise of common ownership, in which the same equity holders simultaneously hold stakes in multiple competing firms. For example, large asset managers such as BlackRock, Vanguard, and State Street have diversified positions across major U.S. airline carriers. This increase in common ownership has important implications for strategic behavior, corporate governance, and market competition across many sectors of the economy. The U.S. airline industry provides an especially salient environment to study these effects, since major domestic carriers operate in concentrated markets, face high fixed costs, and rely heavily on capital investment and technological efficiency to remain competitive.

Much of the existing research focuses on whether common ownership reduces product-market competition. Azar et al. (2018) argue that common ownership weakens price competition in the airline industry, while Dennis et al. (2022) show that the positive correlation

found in prior work can be explained by market concentration rather than ownership structures themselves. While there are numerous empirical studies on the effects of common ownership on pricing and output, much less is known about how common ownership affects firms internally, particularly their productivity, technological investment, and cost efficiency.

Airlines operate in a technologically complex and capital-intensive environment, where efficiency depends on many factors, such as optimizing routes, labor allocation, fleet composition, and fuel usage. Productivity is therefore closely tied to both long-term competitiveness and firm survival. Understanding whether common ownership disincentivizes innovation or cost reduction is essential for interpreting how modern financial markets shape firm behavior.

Economic theory presents contrasting predictions. On one hand, common owners may discourage the aggressive innovation that threatens rival firms in favor of maximizing combined portfolio profits, potentially reducing productivity growth. Alternatively, corporate governance and information spillovers provide mechanisms for institutional investors to facilitate better management and share knowledge between firms in their portfolios, which could increase productivity.

1.2 Research Questions and Hypotheses

This project investigates whether or not common ownership in the U.S. airline industry influences firm-level productivity. Specifically, we address the following:

Research Questions:

1. Does common ownership weaken firms' incentives to invest in cost-reducing technologies or operational improvements?
2. Is common ownership associated with lower firm-level Total Factor Productivity (TFP)?

Based on theoretical predictions that reduced competitive pressure may diminish innovation incentives, we propose two testable hypotheses:

1. Higher common ownership decreases incentives for technological innovation and efficiency improvements.
2. Increases in common ownership are associated with lower Total Factor Productivity.

Based on the pricing literature, these hypotheses follow the logic that if common ownership truly leads firms to internalize competitors' profits, it will weaken both aggressive price competition and productivity investment.

1.3 Contributions and Empirical Approach

To evaluate these hypotheses, we combine quarterly carrier-level operational and financial data from the Department of Transportation's (DOT) Form 41 with institutional ownership data from the U.S. Securities and Exchange Commission's (SEC) Form 13F filings. We estimate

firm-level productivity using the method of Levinsohn and Petrin (2003), which builds on Olley and Pakes (1996) by using intermediate inputs as a proxy for unobserved productivity. This approach is particularly suitable for the airline industry, which exhibits lumpy investment and endogenous input choices. To measure common ownership, our study uses the Modified Herfindahl-Hirschman Index ($MHHI\Delta$), which was also used in Azar et al. (2018) and Dennis et al. (2022). We then examine whether within-firm changes in common ownership correlate with changes in TFP using a fixed effects framework. This empirical design allows us to isolate ownership-driven variation from firm and time trends.

Our contribution has three parts. First, we extend the common ownership literature beyond pricing to study technological efficiency and productivity, a dimension that has no established theoretical framework despite its importance for long-run industry performance. Second, we incorporate Levinsohn-Petrin’s production function estimation techniques to construct a rigorous measure of airline productivity. Third, we provide new descriptive evidence on whether modern ownership structures have reshaped the operational incentives of airlines.

2 Literature Review

2.1 Common Ownership and Anti-Competitive Pricing Behavior

The rise of institutional investment has inspired many empirical papers on whether common ownership influences firms’ competitive behavior. The foundational debate originated with Azar et al. (2018), who argue that overlapping institutional owners reduce product-market competition in the airline industry. Using market-level variation in investor shares and fare prices, Azar et al. (2018) claim that firms with common owners tend to charge higher fares and offer lower output, consistent with anti-competitive incentives.

Dennis et al. (2022) provide a critique of these findings. Their study is also set in the U.S. airline industry and argues that the positive correlation identified by AST does not arise from ownership structure but instead reflects underlying market concentration. Using both carrier-level and market-level variation, DGS demonstrate that once concentration and route fixed effects are controlled for, the relationship between common ownership and fares disappears. They conclude that common ownership does not causally reduce competition or raise prices. Importantly, Dennis et al. (2022) highlight a necessary condition for any anti-competitive effect: there must exist a systematic correlation between ownership metrics (e.g., $MHHI\Delta$) and relevant firm behavior.

While prior work focuses almost exclusively on pricing, neither Azar et al. (2018) nor Dennis et al. (2022) address whether common ownership alters firm productivity, innovation incentives, or long-run cost efficiency. Therefore, the relationship between ownership structure and internal firm performance remains theoretically ambiguous and empirically unexplored, especially in the context of airlines.

2.2 Productivity in the Airline Industry

To understand how ownership might affect productivity, it is important to consider previous research on efficiency dynamics within the airline sector. Semenick Alam and Sickles (2000) provide one of the earliest and most comprehensive analyses of airline productivity. Using panel data from 1970-1990, they estimate technical efficiency through non-parametric methods and report substantial variation in productivity growth across carriers. Their findings show that deregulation and competitive pressures drove efficiency toward convergence. This suggests that airlines respond strongly to changes in market structure, cost shocks, and technological opportunities.

This is relevant to our study because it demonstrates that productivity varies between firms, fostering the motivation to investigate factors such as stakeholder incentives which may influence this dispersion. Additionally, previous studies suggest that changes in competition and environmental shocks have historically shaped airlines' operational efficiency, which implies that ownership-driven changes in strategic incentives may have an effect on productivity outcomes.

2.3 Common Ownership and Productivity

While industry-level airline studies on ownership and productivity are limited, insights from broader cross-country studies help frame potential mechanisms. Bas et al. (2023) examine common ownership and productivity across 18 OECD countries. Their findings highlight important heterogeneity: inter-industry common ownership is associated with high productivity, possibly due to better coordination or information spillovers, while intra-industry common ownership tends to have negative and insignificant effects on productivity, consistent with weakened competition or disincentivized innovation. Although the airline industry is not the focus of their study, the findings of Bas et al. (2023) are relevant to the theoretical context of our analysis. Specifically, they emphasize the need to empirically evaluate whether the governance or common ownership channel dominates for domestic carriers.

2.4 Measuring Productivity and Efficiency

Accurately measuring productivity is essential for our research question. There are two strands of literature that provide the framework for modern production function estimation. First, Olley and Pakes (1996) propose an estimator that corrects for simultaneity and selection bias. Simultaneity bias occurs when firms choose inputs based on unobserved productivity shocks, and selection bias arises from endogenous firm exit. The second was proposed by Levinsohn and Petrin (2003), which builds upon Olley and Pakes' method by using intermediate inputs, instead of investment, as a proxy for unobserved productivity. This modification improves estimation in industries where investment is lumpy, infrequent, or zero. Given the irregular investment patterns of airlines, this method is particularly well suited for our quarterly carrier-level dataset. While these papers do not address ownership or competition directly, they provide the methodological foundation for our TFP estimation.

2.5 Gaps in the Literature

Although prior research highlights several important points related to our study, a clear gap remains. Theory provides multiple channels through which common ownership could either enhance or deter productivity, but no empirical study has tested these predictions for U.S. airlines. Our project directly addresses this gap by integrating ownership measures from SEC Form 13F data with production function estimates of TFP, allowing us to explore whether overlapping ownership structures correlate with firms' technological efficiency.

3 Methodology

This section details our empirical strategy for estimating the effect of common ownership on firm-level productivity in the American airline industry. The analysis is carried out in three stages. First, we apply the methodology proposed by Levinsohn and Petrin (2003) to estimate firm-level Total Factor Productivity, the proxy variable for technological innovation and efficiency. Second, we define and calculate the degree of common ownership using the Modified Herfindahl-Hirschman Index ($MHHI\Delta$) at the carrier level as described by Azar et al. (2018). Finally, we describe the model specification used to test our hypothesized relationship between estimated TFP and $MHHI\Delta$.

3.1 Estimating Total Factor Productivity

Following Levinsohn and Petrin (2003), we begin by assuming a standard Cobb-Douglas production function. Applying a log-transform yields the following specification:

$$\ln(Y_{it}) = \beta_l \ln(L_{it}) + \beta_k \ln(K_{it}) + \omega_{it} + \varepsilon_{it} \quad (1)$$

where Y_{it} is real output, L_{it} is real labor input, K_{it} is real capital input, ω_{it} is the unobserved productivity term, and ε_{it} is the error term. Given this model specification, three estimation challenges arise. First, a firm's choices of labor, L_{it} and capital K_{it} are dependent upon each other, leading to an endogeneity concern. Capital can be considered a semi-fixed input, thus a firm's choice of capital may depend on labor in a dynamic sense (they may choose to invest in capital now, knowing payoff will come later). As a result, per period capital is treated as fixed by the firm and thus is independent of labor. The second econometric concern is that a firm's productivity ω_{it} is correlated with a firm's choice of labor and capital, leading to a simultaneity bias. Correcting for this bias is the major contribution by Olley and Pakes (1996), and later Levinsohn and Petrin (2003), which we employ next. The third and final challenge is the survival bias of productive firms. That is, a firm that is less productive is more likely to leave the market, biasing productivity estimates. We consider the latter two concerns in sequence.

In order to correct for the simultaneity bias produced when naively estimating 1, Levinsohn and Petrin (2003) suggest the following. First, suppose we can model productivity ω_{it} as a non-parametric function of our semi-fixed input capital K_{it} and an intermediate input M_{it} such

as material costs:

$$\omega_{it} = \varphi(\ln(K_{it}), \ln(M_{it})) \quad (2)$$

If we suppose that the intermediate input $\ln(M_{it})$ is monotonically increasing in ω_{it} , which follows from the assumption that a more productive firm will use more materials, we can guarantee this inversion exists. More formally, given the following materials demand equation,

$$\ln(M_{it}) = h(\omega_{it}, \ln(K_{it})) \quad (3)$$

the implicit function theorem guarantees that 3 is locally invertible, so we can uniquely determine some nonparametric function 2. Using this correction, we next modify our model specification to correct for this simultaneity bias.

Given our function 2, we can substitute this expression in the production function 1 yielding:

$$\ln(Y_{it}) = \beta_l \ln(L_{it}) + \beta_k \ln(K_{it}) + \varphi(\ln(K_{it}), \ln(M_{it})) + \varepsilon_{it} \quad (4)$$

and defining

$$\Phi(\ln(K_{it}), \ln(M_{it})) = \beta_k \ln(K_{it}) + \varphi(\ln(K_{it}), \ln(M_{it})) \quad (5)$$

Thus we can define the following first-stage estimation equation,

$$\ln(Y_{it}) = \beta_l \ln(L_{it}) + \Phi(\ln(K_{it}), \ln(M_{it})) + \varepsilon_{it} \quad (6)$$

Hence, we are now able to recover an unbiased estimate of our labor coefficient β_l . However, our coefficient for capital β_k has not yet been identified.

To identify the coefficient for capital (β_k) and produce accurate estimates of ω_{it} , Levinsohn and Petrin (2003) introduces the following second stage. First, assume productivity follows a first-order Markov process:

$$\omega_{it} = g(\omega_{i,t-1}) + \xi_{it} \quad (7)$$

where our function g is unknown and ξ_{it} is a productivity shock. This step establishes that a firm chooses labor L_{it} and capital K_{it} based on $\omega_{i,t-1}$ as they cannot anticipate ξ_{it} . Hence, decisions in period t are conditional on $\omega_{i,t-1}$ but not on ξ_{it} , which gives us the moment condition required to estimate our capital coefficient β_k .

Using our first-stage estimation of $\hat{\Phi}$ we can estimate productivity,

$$\hat{\omega}_{it} = \ln(Y_{it}) - \beta_l \ln(L_{it}) - \hat{\Phi}(\ln(K_{it}), \ln(M_{it})) \quad (8)$$

and then estimate the function g by regressing $\hat{\omega}_{it}$ on $\hat{\omega}_{i,t-1}$. So the difference between observed and predicted productivity yields innovation

$$\hat{\xi}_{it} = \hat{\omega}_{it} - \hat{g}(\hat{\omega}_{i,t-1}) \quad (9)$$

which is uncorrelated with capital K_{it} by construction. Hence the moment conditions

$$\mathbb{E} \left[\hat{\xi}_{it} \cdot K_{it} \right] = 0 \quad (10)$$

allow us to solve for β_k and thereby recover the productivity estimates.

Having corrected for the simultaneity bias, we must now correct for the selection bias of more productive firms. This bias arises because a firm's decision to exit the market is endogenous to its unobserved productivity. Thus less productive firms are more likely to fail and drop out of the sample, meaning remaining observations are not randomly selected and will bias estimated market-level productivity averages towards more productive firms, meaning that standard estimation techniques would misrepresent the true relationship. To correct for this Olley and Pakes (1996) explicitly model the survival decision. The survival decision is modeled as a function of the firm's state variables, capital (K_{it}) and unobserved productivity (ω_{it}) as:

$$\mathcal{P}_{it} = \Pr(\chi_{it} = 1 | \cdot) = \mathcal{P}_t(K_{it}, \omega_{it}) \quad (11)$$

Where χ_{it} is an indicator variable equal to one if the firm continues operation at time t . The firm remains in the market if its expected future profit is non-negative, which depends on ω_{it} . However, the LP estimation model is designed to provide consistent estimates of β_l and β_k by implicitly accounting for this selection rule. Specifically, the second-stage moment condition is only estimated over the subset of firms that survive. The use of the control function approach, where the parameter $\Phi(\cdot)$ captures the relationship between ω_{it} and the variable inputs ensures that the estimated input elasticities are consistent. By assuming that the functional form for survival probability $\mathcal{P}_{it}$ is monotonically increasing in ω_{it} (more productive firms are more likely to survive), the two-stage procedure generates consistent production function estimates that are robust to this selection bias.

Using this procedure, we recovered carrier-level estimates of productivity, see subsection 5.1 for estimation results.

3.2 Measuring Common Ownership

The second critical component of our empirical strategy is producing an estimate of firm-level degree of common ownership. Following from established literature, we use the *MHHIΔ* (Modified Herfindahl-Hirschman Index) developed by Azar et al. (2018) and utilized in Dennis et al. (2022). This metric measures the extent to which a firm and its competitors are owned by the same institutional investor. This serves effectively as a proxy for the investors' incentives to encourage non-competitive behavior.

The *MHHIΔ*, for a firm (airline) i at time t captures the aggregate commonality of ownership between the firm i and all of its competitors j , with weighting by market shares. Formally, it is defined as:

$$MHHI\Delta = \sum_{j \neq i}^N \left(s_{jt} \cdot \sum_k \alpha_{ikt} \alpha_{jkt} \right) \quad (12)$$

Where s_{jt} is the market share of the rival firm j , N is the number of firms in the market, k is

the index of each institutional investor, and the term $\sum_k \alpha_{ikt} \alpha_{jkt}$ represents the commonality of ownership between firm i and a single rival j .

Similar to the standard Herfindahl-Hirschman Index (HHI), a higher value of the $MHHI\Delta$ implies a higher degree of commonality among competitors. We hypothesize that an increase in $MHHI\Delta$ reduces competitive pressures. In line with Dennis et al. (2022) and Azar et al. (2018), we consider the interaction of $MHHI\Delta$ with competitive pressures, however we consider how $MHHI\Delta$ interacts with competitive pressures for cost reduction and innovation. Specifically, if common ownership discourages a firm from investing in new technologies that would significantly reduce its marginal cost, giving it a competitive advantage, it will lead to a decrease in total factor productivity (TFP). Thus, our main hypothesis postulates the following relationship,

$$\uparrow MHHI\Delta \implies \downarrow TFP \quad (13)$$

Thus our model specification will focus on establishing this relationship.

3.3 Empirical Model Specification

The final stage of our methodology establishes the model specification we employ to estimate the effect of common ownership, as measured by $MHHI\Delta$, on our estimated total factor productivity, $\hat{\omega}_{it}$. This subsection introduces our model specification as well as provides a brief justification.

We model the estimated carrier-level productivity $\hat{\omega}_{it}$ as a function of our primary explanatory variable $MHHI\Delta_{it}$, and controls, as follows:

$$\hat{\omega}_{it} = \beta_0 + \beta_1 MHHI\Delta_{it} + \gamma X_{it} + \mu_i + \delta_t + \varepsilon_{it} \quad (14)$$

Where, $\hat{\omega}_{it}$ is our estimated productivity term, $MHHI\Delta_{it}$ is our measure of common ownership and our primary explanatory variable, X_{it} is a vector of control variables, and μ_i and δ_t are firm and time fixed effects respectively. In line with previous literature, namely Dennis et al. (2022) and Azar et al. (2018), we are sure to control for firm size.

4 Data and Variables

This section describes the data sources, variable construction, and summary statistics used in our analysis of common ownership and airline productivity. The discussion proceeds in four parts. First, we introduce the airline operational and financial data drawn from DOT Form 41. Second, we describe the institutional ownership data from SEC Form 13F filings. Third, we define the construction of our key inputs for TFP estimation (outputs, labor, capital, and materials) and the inputs used to compute common ownership (market shares and ownership stakes). Finally, we present descriptive statistics for the main variables used in the empirical analysis.

4.1 Airline and Financial Data Sources

Carrier-level operating and financial variables are obtained from the U.S. Department of Transportation (DOT) Form 41 database, which provides standardized quarterly disclosures for U.S. air carriers. We focus on domestic passenger carriers and construct a quarterly panel over our analysis window (Q2 2013 to Q4 2017), yielding $N = 767$ carrier-quarter observations in the main sample.

Form 41 schedules are used to construct the production inputs and output employed in the Levinsohn and Petrin (2003) procedure. Following the measurement conventions in the airline productivity literature (e.g., Semenick Alam and Sickles (2000)), we map Form 41 accounts into the four required components:

1. **Output:** Operating revenue from Form 41 Schedule P-1.2.
2. **Labor:** Salaries and fringe benefits from Form 41 Schedule P-6.
3. **Capital:** Aircraft and flight equipment (capital stock proxy) from Form 41 Schedule B-1.
4. **Materials:** Domestic fuel cost from Form 41 Schedule P-12(a).

4.2 Ownership Data

Institutional ownership is measured using SEC Form 13F filings. Form 13F reports quarterly equity holdings by institutional investment managers with assets valued over one hundred million dollars. We identify airline issuers using CUSIP identifiers and map each issuer to a carrier code. For each carrier i , investor k , and quarter t , we observe holdings (shares and reported value), which we use to construct ownership shares and common ownership measures.

A key practical challenge is that Form 13F filings cover institutional investors rather than the full universe of shareholders. Consistent with prior empirical work, we treat the 13F universe as capturing the relevant set of diversified investors for common ownership incentives, and we construct ownership shares using the observed institutional holdings. In the implementation used for our baseline common ownership measure, we proxy total shares outstanding within a carrier-quarter by the total number of shares held by all investors in the 13F sample for that carrier-quarter. This produces an internally consistent ownership share for each investor:

$$\alpha_{ijt} \equiv \frac{\text{shares}_{ijt}}{\sum_k \text{shares}_{ikt}}.$$

We use these ownership shares to construct pairwise “profit weights” across carriers (as in Dennis et al. (2022)) and then aggregate them into a carrier-level common ownership pressure measure (described below).

4.3 Variable Construction and Deflation

All nominal DOT Form 41 variables are converted to real values using deflators chosen to match the underlying revenue or cost category. Specifically, we construct:

- **Real Output** (Y_{it}): operating revenue (Schedule P-1.2), deflated by the PPI for Scheduled Passenger Air Transport.
- **Real Capital** (K_{it}): aircraft equipment (Schedule B-1), deflated by the PPI for Transportation Equipment: Aircraft.
- **Real Labor** (L_{it}): salaries and fringe benefits (Schedule P-6), deflated by Average Hourly Earnings for Transportation Workers.
- **Real Materials** (M_{it}): domestic fuel cost (Schedule P-12(a)), deflated by the PPI for Jet Fuel.

We then take natural logs of the resulting real variables to form $\ln(Y_{it})$, $\ln(L_{it})$, $\ln(K_{it})$, and $\ln(M_{it})$, which are the inputs to the Cobb–Douglas production function in equation (1) and the Levinsohn–Petrin estimation procedure described in Section 3.1.

To construct our common ownership metric, we require (i) carrier market shares and (ii) institutional ownership stakes. We define each carrier’s market share s_{jt} at time t using that carrier’s share of total domestic activity in the quarter (in our baseline implementation, the carrier’s share of total domestic operating revenue). Institutional ownership shares α_{ikt} are obtained from SEC Form 13F filings and represent the fraction of carrier i owned by institutional investor k in quarter t . Combining these objects, we compute carrier-level common ownership using the Modified Herfindahl–Hirschman Index delta, $MHHI\Delta_{it}$, exactly as defined in equation (12):

$$MHHI\Delta_{it} = \sum_{j \neq i} \left(s_{jt} \cdot \sum_k \alpha_{ikt} \alpha_{jkt} \right). \quad (15)$$

Because our analysis is conducted at the carrier (rather than route-market) level, s_{jt} is constructed at the national domestic level for each quarter. This choice is dictated by the availability of carrier-level production data required for TFP estimation, and it implies that the relevant “market” for computing $MHHI\Delta_{it}$ is the set of domestic carriers in period t .

Finally, we construct the additional controls used in the second-stage regression in equation (14). In line with Dennis et al. (2022) and Azar et al. (2018), our baseline X_{it} includes a measure of firm size (proxied by $\ln(K_{it})$) and a bankruptcy indicator capturing quarters in which the carrier operated under Chapter 11 protection. Where included, we measure market concentration using the standard index $HHI_t \equiv \sum_j s_{jt}^2$, constructed from the same carrier-level market shares s_{jt} .

4.4 Descriptive Statistics

Table 1 reports descriptive statistics for our estimated TFP measure. For the full set of results, see Appendix B. These summary measures indicate substantive but not extreme heterogeneity in firm-level productivity within our sample.

Table 1: Descriptive Statistics: Estimated TFP

Statistic	TFP ($\hat{\omega}_{it}$)
Observations (N)	767
Mean	1.224
Median	1.216
Std. Dev.	0.182
Skewness	0.157
5th percentile	1.023
95th percentile	1.580

Table 2 also summarizes the common ownership measure ($MHHI\Delta$) and the control variables (e.g., $\ln(K_{it})$, bankruptcy indicator). Because common ownership is computed at the carrier-quarter level, its variation reflects both changes in institutional holdings and changes in carrier size (market share). Consistent with the motivation of Azar et al. (2018) and Dennis et al. (2022), higher values of $MHHI\Delta$ indicate greater overlap in ownership between a carrier and its rivals, implying potentially weaker incentives for aggressive competition.

Table 2: Descriptive Statistics: Controls

Variable	N	Mean	Std. Dev.	5%	95%
$MHHI\Delta$	196	0.398	0.114	0.251	0.615
$\ln(K_{it})$	196	15.2	1.76	12.3	17.2
Bankruptcy Dummy	196	0.031	0.173	N/A	N/A

Note: $MHHI\Delta$ is computed from SEC 13F holdings and revenue-based market shares; $\ln(\text{Capital})$ is the log of real aircraft equipment; Bankruptcy Dummy equals 1 during Chapter 11 protection quarters.

5 Empirical Results

This section presents the empirical results in three steps. First, we summarize the results from the Levinsohn and Petrin (2003) productivity estimation procedure. Second, we report estimates from the main fixed effects regression of productivity on common ownership, as specified in equation (14). Third, we interpret the lack of statistical significance for $MHHI\Delta$ and relate the findings to the conclusions in Dennis et al. (2022).

5.1 TFP Estimation Results

Using the methodology described in Section 3, we estimate firm-level total factor productivity from a Cobb–Douglas production function with a fuel-based proxy for unobserved productivity shocks. In keeping with the structure of airline production, labor is treated as a free input and

capital as a semi-fixed input, while fuel expenditures provide the intermediate input required for the Levinsohn–Petrin inversion.

The estimated input elasticities for labor and capital are economically reasonable and consistent with the airline productivity literature. In particular, the labor elasticity is estimated precisely and accounts for a substantial portion of short-run variation in output, while the meaningful role of capital reflects the importance of fleet and equipment in generating revenue. Consistent with the discussion of simultaneity bias in Section 3, naive estimation approaches typically understate the contribution of capital; the LP procedure is designed to correct for this bias through the proxy-based identification strategy.

We recover the productivity term $\hat{\omega}_{it}$ for each carrier-quarter and use it as our primary dependent variable in the second-stage regressions. The distribution and summary statistics of $\hat{\omega}_{it}$ are reported in Table 1, for full statistics see Appendix A, B, & D.

5.2 Main Regression Results

We estimate the effect of common ownership on productivity using the fixed effects model in equation (14). The baseline regression includes carrier fixed effects (μ_i), year fixed effects (δ_t), and a set of controls. Standard errors are clustered at the carrier level.

Table 3 reports the highlighted results; see Appendix Table 6. The coefficient on MHIA is negative (-0.036), consistent in sign with the hypothesis that increased common ownership could reduce competitive incentives for cost reduction or innovation. However, the estimate is statistically insignificant ($p\text{-value} = 0.415$), implying that within-carrier changes in common ownership are not systematically associated with changes in estimated productivity in our data.

The control variables behave as expected. The bankruptcy dummy is negative and statistically significant, indicating that carriers in financial distress exhibit lower measured productivity. The coefficient on $\ln(\text{Capital})$ is negative and significant. One interpretation is that the survivor bias is not fully accounted for biasing the estimate down, as a larger firm can survive a poor productivity shock. The year fixed effects indicate an upward trend in productivity over the sample period relative to the base year 2013. This is consistent with typical productivity growth over time.

5.3 Discussion of Non-Significant Findings

Our baseline specification fails to reject the null hypothesis of no relationship between common ownership and firm-level productivity. Interpreting this result requires caution: the estimate does not imply that common ownership cannot affect innovation incentives, but rather that in this setting and with the available carrier-level variation, we do not detect a statistically meaningful association between MHIA and $\hat{\omega}_{it}$.

This finding closely parallels the core message in Dennis et al. (2022). Dennis et al. emphasize that empirical claims about anti-competitive outcomes require, at minimum, evidence of correlation between common ownership measures and competitive behavior. In their pricing context, they show that once concentration-related components of common ownership metrics are appropriately controlled for, the remaining correlation with prices largely disappears. Our

Table 3: Main Regression Results

Productivity ($\hat{\omega}_{it}$)	Coefficient	p-value
MHHI Δ	-0.036	0.415
ln(Capital)	-0.068***	0.000
Bankruptcy Dummy	-0.056***	0.000
Time Fixed Effects (Base Year: 2013)		
Year 2014	0.012*	0.091
Year 2015	0.028***	0.001
Year 2016	0.050***	0.002
Year 2017	0.068***	0.002

Standard errors are clustered at the carrier level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

results suggest a similar conclusion for productivity: after accounting for carrier fixed effects, time effects, and key controls, changes in common ownership are not correlated with systematic changes in measured productivity.

6 Further Analysis and Robustness

The primary empirical finding of this analysis, the lack of a statistically significant correlation between common ownership, $MHHI\Delta$, and productivity $\hat{\omega}_{it}$, does not definitively prove that common ownership has no effect on productivity. Rather, this suggests that the effect, if at all present, is not captured by our model specification.

Following from the fact that a correlation must exist for any causal effect to be present, this non-finding compels a further exploration of the two primary possibilities that could explain the lack of estimated significance, first that our model is misspecified, and second we consider alternative causal explanations. This section will investigate these two avenues in sequence.

6.1 Possible Model Misspecification

The first potential explanation for the statistically insignificant estimate is that the current model specification is misspecified. This suggests that there exists a correlation between common ownership and productivity, but it is obscured or biased by limitations of this specification. These limitations fall into three categories: omitted variable bias, incorrect dynamic assumptions, or inadequate functional forms, each of which we will address.

Omitted Variable Bias: A critical source of potential misspecification is the failure to include relevant variables in the controls (X_{it}) that influence productivity and are simultaneously correlated with common ownership. This error leads to OVB and causes estimates to be biased or inconsistent. In our context, we must ensure that the residual variations in productivity are not being attributed to common ownership when they are being driven by unobserved or uncontrolled firm characteristics. We identify the following as key potential sources of OVB.

1. **Financial Health;** A key characteristic of any firm is the firm’s financial conditions. A firm in poor financial health will struggle to purchase materials, fully utilize capital, or properly compensate workers, meaning that while the firm’s productivity may not be impacted, its ability to utilize this productivity may be. Thus, using a variable such as Debt-to-Equity ratios, or other indicators of financial health, will allow for a clearer image of true productivity. Further, institutional investors often consider the financial health of a firm before investing, meaning the financial conditions of a firm are likely correlated with the level of commonality of a firm.
2. **Technological Lag;** In many industries, particularly the airline industry, new technologies often take time to be adopted. Of primary concern in the airline industry is fleet age, which is a direct proxy for the technological lag of a firm’s capital stock. Newer aircraft are significantly more fuel-efficient, require less maintenance, as well as other superior technologies, all of which contribute directly to a firm’s productivity. Firms with greater financial flexibility may be better positioned for large capital expenditures on new aircraft, and firms with institutional investors often benefit from the deep pockets of these investors, allowing them to make these larger capital expenditures.
3. **Spatial and Specific Fixed Effects;** While our model already includes a global firm and time fixed effects variable, these variables only capture global variations. However, certain competitive or operational conditions may vary by region, and are not fully absorbed by the general firm fixed effect, but have time-varying impact on productivity. Thus, if common ownership is consistently higher in airlines that dominate specific regional market or operational clusters, and those groups experience differential productivity trends, omitting a regional control would lead to OVB.

Incorrect Dynamic Assumptions: Our current model operates under the assumption that any changes in common ownership have an instantaneous effect on firm level productivity, within the same quarter. In reality the anti-competitive effects of common ownership on productivity are likely to exhibit significant lag. There are several reasons why the effect on productivity may not be same-quarter.

First, while an investor immediately gains control over a firm upon purchasing shares, an investor’s decisions that discourage competitive innovation must first be translated into a change in corporate policy or resource allocations. These strategic shifts require time to draft and implement, then propagate through the firm. Second, the impact of an internal strategic change on productivity may not be immediately visible in output but will emerge gradually. Finally, since a large component of technological innovation involves capital investment (e.g., new aircraft or maintenance equipment), the inherent stickiness and long lead times associated with capital acquisition mean that the full effect of reduced investment will be distributed across several future periods.

Inadequate Functional Forms: The third potential source of misspecification we address is our assumption of linearity of ownership and productivity. Previous works have suggested that the influence of common ownership may be nonlinear, implying the existence of a critical mass or threshold effect.

At low levels of common ownership, the effects on corporate strategy are likely to be minuscule or entirely negligible. Institutional investors with very small stakes in rival firms lack the incentive or power to influence management decisions regarding competitive investment. As institutions amass ownership, they gain sufficient incentive to facilitate anti-competitive behavior, as well as voice to pressure management, either directly through shareholder votes or indirectly through their collective influence on capital. Thus, a disincentive on innovation may only take place at large levels of common ownership.

6.2 Alternative Causal Explanations

The second possibility explaining the statistically insignificant correlation is that the lack of correlation is genuine, and that common ownership and productivity are uncorrelated. This scenario aligns with findings of DGS regarding pricing behavior, where market concentration, not common ownership, was the primary driver of anti-competitive pricing. If common ownership is indeed causally irrelevant to productivity, we must investigate why the theoretical link breaks down in the airline industry.

This subsection will explore the theoretical reasons why common ownership does not affect firm level productivity. We will focus on factors related to investor behavior, distinguishing between price and cost competition, and potential beneficial effects of common ownership.

Investor Passivity: One significant factor that may separate common ownership from changes in firm productivity is the prevalence of investor passivity among large institutional shareholders. The $MHHI\Delta$ metric assumes that common ownership implies common voice, or a coordinated strategic influence designed to dampen competition. However, this assumption is often inaccurate given the nature of modern institutional investment.

Index funds drive much of the common ownership, but these funds are legally and strategically obligated to hold shares in all constituent firms of an index, regardless of competitive relationships between firms. In combination with the minimal incentives for fund managers of these passive funds to engage in detailed corporate governance in regard to decisions affecting productivity, it is likely that while shares may be commonly held, there is no voice behind these shares.

If a significant portion of the $MHHI\Delta$ is held by passive investors then their ownership stakes may not translate into effective control or anti-competitive incentives concerning productivity. A passive investor may be content for a firm to continue competing fiercely on cost efficiency. Therefore, we may overestimate the true extent of common influence.

Price vs. Cost Competition: The theory of common ownership primarily focuses on the incentive to reduce price competition, as this directly increases margins across all commonly-held firms. However, this does not necessitate that cost competition be eliminated or affected. Firms facing common ownership may still choose to compete vigorously on the cost efficiency dimension for several different reasons.

First, a firm that successfully implements cost savings sees its own profits rise. Even if the firm's common owners benefit from higher rival prices, they still maximize their portfolio returns when each individual firm is as profitable as possible. Cost reduction is a firm-specific

good, whereas price collusion is a cross-firm good.

Second, if a firm does not invest in cost reduction, it risks becoming structurally inefficient compared to rival firms. This inefficiency makes the firm vulnerable to market share loss if tacit pricing agreements break down, or if non-commonly owned rivals enter the market. Therefore, the threat of future competition can sustain the incentive for current investment in productivity.

Finally, management compensation is often tied to firm-specific performance metrics rather than industry-wide pricing. Management, seeking to maximize their own utility, retains a strong incentive to push for technological innovation, even if common owners would prefer reduced price competition. Thus this divergence means that common ownership may be causal for price collusion but not for productivity collusion.

Beneficial Effects: The final causal explanation we propose is that, even if common ownership exerts the hypothesized pressure to reduce cost competition, this negative influence may be offset by or dominated by simultaneous, opposing productivity forces that arise from common ownership. This means common ownership may be a double-edged sword, where its benefits equal or outweigh its drawbacks.

Two primary channels exist through which common ownership could increase firm-level productivity. First, institutional ownership often acts as a credible signal of a firm’s long-term stability and quality, so a higher degree of common ownership may signal the broader market that the firm has stable governance. This perception of reduced risk can lower a firm’s cost of capital. A lower cost of capital enables a firm to undertake more capital-intensive, high-risk, technological investments that may boost productivity.

The second channel by which common ownership may boost productivity is shared best practices and governance. Common ownership provides a governance structure that spans across multiple firms. This structure may facilitate the sharing or rapid adoption of non-competitive technological innovations and best practices that are industry-wide public goods. This accelerated diffusion of efficiency gains could lead to a net increase in productivity across the industry, potentially masking any deliberate slowdown in productivity growth.

7 Conclusion

7.1 Summary of Findings

This paper investigates whether common ownership influences firm-level productivity in the U.S. airline industry. Drawing on quarterly operational and financial data from DOT Form 41, ownership data from SEC 13F filings, and productivity estimates derived using the Levinsohn-Petrin (2003) method, we examine whether common institutional investors alter carriers’ incentives to innovate or reduce costs. Dennis, Gerardi, and Schenone (2022) highlight the necessary condition for anticompetitive effects is that there must exist a detectable correlation between ownership and firm behavior. Our study leverages this condition to test whether common ownership is systematically related to Total Factor Productivity.

Across all specifications, our results show no statistically significant relationship between the common ownership measure ($MHHI\Delta$) and estimated productivity. The coefficient for

$MHHI\Delta$ is small and insignificant, and the sign is not robust across alternative controls. Although our TFP estimation reproduced expected relationships for key inputs (e.g., capital and labor), the ownership variable itself shows no meaningful association with technological efficiency. Importantly, the absence of correlation does not establish evidence against a causal effect but instead highlights the empirical difficulty of identifying such effects with available carrier-level data.

7.2 Policy Implications

Our findings mirror a broader trend in common ownership literature. As DGS (2022) argue, the observed relationship between common ownership and pricing behavior is largely driven by market concentration rather than direct ownership incentives, our analysis suggests that common ownership may also be unrelated to productivity dynamics in domestic airlines. If overlapping ownership does not meaningfully distort firms’ internal incentives for innovation or cost reduction, then the concerns raised in the pricing literature may not extend to productivity.

From a policy perspective, the results underscore the complexity of evaluating common ownership through a single empirical lens. Regulators often rely on ownership concentration metrics such as $MHHI\Delta$ when assessing potential anticompetitive effects. Yet our study suggests that these metrics, while informative, may be insufficient for understanding nuanced operational outcomes like efficiency or technological innovation. Any regulatory actions targeting common ownership would require a deeper understanding of firm-level behavior, investor engagement, and industry-specific innovation trends.

7.3 Avenues for Future Research

Although our model provides a first step toward studying common ownership and productivity, several limitations prevent us from drawing causal conclusions. The null result of this study may reflect issues with model specification rather than the absence of any true relationship. A promising direction for future work is to develop a formal theoretical framework linking ownership structure to productivity incentives. This would enable a more targeted empirical design by clarifying which mechanisms, such as reduced competition, governance spillovers, or capital-market effects, should matter in theory. Future research could also incorporate richer data, including route-level productivity measures, aircraft-level technological features, or investor engagement proxies, to better isolate the channels through which common ownership might affect firm performance.

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A Appendix: TFP Estimation Results

Table 4: Production Function Estimation (Levinsohn-Petrin Method)

Dependent Variable:		ln(Revenue)
Variable	Coefficient	Std. Error
ln(Labor)	0.8629***	(0.0808)
ln(Capital)	−0.0155	(0.0450)
ln(Fuel)	0.1032**	(0.0426)
Estimation Summary		
Estimation Method	Levinsohn-Petrin (LP)	
Production Function	Cobb-Douglas	
Observations (N)	767	
Number of Groups (J)	46	
χ^2 (Constant Returns Test)	0.22	

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

The test for Constant Returns to Scale has a null hypothesis that the sum of the input elasticities ($\beta_{\text{Labor}} + \beta_{\text{Capital}} + \beta_{\text{Fuel}}$) equals 1. The high p -value (0.64) suggests that the hypothesis cannot be rejected.

B Appendix: TFP Distribution Details

Table 5: Detailed Descriptive Statistics: Total Factor Productivity (TFP)

Percentiles		Distribution Statistics	
1%	0.8841	Observations	767
5%	1.0234	Variance	0.0332
10%	1.0439	Skewness	0.1567
25% (Q1)	1.1145	Kurtosis	3.4797
50% (Median)	1.2155	Minimum	0.3486
75% (Q3)	1.3990	Maximum	1.9448
90%	1.5234		
95%	1.5559		
99%	1.6339		

C Appendix: Full Regression Tables

Table 6: Regression on Total Factor Productivity ($\hat{\omega}_{it}$)

Dependent Variable:		$\hat{\omega}_{it}$
Variable	Coefficient	Robust Std. Error
MHHIΔ (CO_Pressure)	−0.0469	(0.0589)
ln(Capital)	−0.0677***	(0.0029)
Bankruptcy Dummy	−0.0554***	(0.0095)
Year Fixed Effects		
2014	0.0123	(0.0066)
2015	0.0325***	(0.0066)
2016	0.0555***	(0.0129)
2017	0.0734***	(0.0181)
Constant	2.4320***	(0.0499)
Estimation Summary		
Observations (N)		192
Number of Groups (J)		11
R ² (Within)		0.8257

Standard errors are clustered by unique carrier (11 clusters).

***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Year 2013 is the omitted base year for the time dummies.

D Appendix: Histogram of Productivity Distribution

